

Statistical Preliminaries

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Chapter 2 of the thesis

Orientation

The methods developed in the following chapters draw on a handful of statistical ideas that, while standard in statistics and machine learning, are not all part of the routine toolkit of a working psychologist. This chapter collects them in one place and explains them at the level of detail needed to follow the rest of the thesis. It is deliberately self-contained and tutorial in tone: a reader comfortable with these concepts may skim it, while a reader who last met some of them in an undergraduate methods course should find here enough to read Chapters 3 to 5 without consulting outside references.

The selection is driven by what the method actually uses. The correlation coefficient (Section) is the raw material of the ReReReRe detector, which compares a respondent’s *individual-level* coherence with a baseline. Exploratory factor analysis (Section) supplies the latent-variable model that justifies why attentive responses to different items covary in the first place, and parallel analysis—the procedure ReReReRe uses to estimate the number of factors—is a factor-retention method. Permutation tests (Section) are the inferential engine behind ReReReRe’s per-respondent z-score, and the bootstrap (Section) provides the confidence intervals used to quantify the detector’s contribution. The Random Forest (Section) is the classifier that combines ReReReRe with the auxiliary detectors into an ensemble, cross-validation (Section) is how its performance is estimated honestly, and the battery of evaluation metrics (Section)—above all the Matthews correlation coefficient—is how detection quality is reported throughout.

Correlation and Covariance

Two variables are *correlated* when knowing one tells us something about the other. The workhorse measure of linear association is the Pearson product-moment correlation coefficient (Pearson, 1896). For two variables X and Y measured on n units, it is defined as the covariance of X and Y divided by the product of their standard deviations,

$$r_{XY} = \frac{\text{cov}(X, Y)}{s_X s_Y} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}. \quad (1)$$

The coefficient ranges from -1 (perfect negative linear association) through 0 (no linear association) to $+1$ (perfect positive linear association). The numerator, the *covariance*, is positive when above-average values of X tend to co-occur with above-average values of Y ; dividing by the standard deviations rescales this to the unit interval, making correlations comparable across variables measured on different scales.

Three properties of Equation 1 matter for what follows. First, r captures only *linear* association: two variables can be strongly related in a curved pattern yet have $r \approx 0$. Second, r is invariant to separate linear rescalings of X and Y : adding a constant to every value, or multiplying every value by a positive constant, leaves r unchanged. This invariance is exploited directly in Chapter 3, where it guarantees that dividing every item by its scale maximum—a rescaling used to handle mixed-format questionnaires—cannot change the correlation among items measured on a common scale. Third, the *sign* of r encodes the direction of the relationship, which is why reverse-keyed questionnaire items (worded so that agreement indicates *less* of the trait) correlate *negatively* with same-trait items keyed in the usual direction; handling that sign correctly is essential when aggregating correlations across item pairs.

A distinction that is central to this thesis, but easy to overlook, is between correlation computed *across respondents* and correlation computed *within a respondent*. The familiar use of Equation 1 treats each respondent as one unit and each pair of items as the two variables X and Y ; this *sample-level* correlation describes how two items relate in the population. One can instead fix a single respondent and treat their answers to a set of item pairs as the data, computing a correlation that describes whether *that person* answered coherently across pairs that the sample says should agree. This *individual-level* or *person-level* correlation is the quantity ReReReRe estimates and tests; the psychometric-synonym index of Chapter 1 is an early instance of the same idea (Meade & Craig, 2012).

Likert Scales and the Measurement Model

Most of the data this thesis concerns come from *Likert items* (Likert, 1932): statements to which respondents indicate agreement on an ordered set of categories, typically five or seven points from “strongly disagree” to “strongly agree.” Strictly, a Likert response is *ordinal*—the categories are ordered, but the psychological distance between “agree” and “strongly agree” need not equal that between “neutral” and “agree.” In practice, researchers routinely treat summed or averaged Likert responses as *interval* data and apply correlation- and mean-based statistics to them, a simplification that is generally defensible for scales with five or more categories that are reasonably symmetric.

The thesis follows this common practice, computing Pearson correlations on Likert responses, with two consequences worth flagging now. First, because the correlation in Equation 1 is sensitive to the numerical spacing of the response options, items administered on *different* Likert ranges (say, some on a 1–5 scale and others on a 1–7 scale) can contribute unequally to an individual-level correlation; Chapter 3 addresses this with a rescaling step. Second, a respondent who selects the same category for every item has zero variance, which makes the denominator of Equation 1 undefined for that person—a degenerate case that careless “straightlining” produces and that any individual-level method must handle explicitly.

Latent Variables and Factor Analysis

Why should answers to different items correlate at all? The dominant answer in psychometrics is the *common-factor model*, which traces to Spearman (1904) and was systematized by Thurstone (1947). The model posits that the observed responses are gen-

erated by a small number of unobserved *latent factors*—traits such as extraversion or conscientiousness—plus item-specific noise. Items that load on the same factor correlate because they share that common cause; items on different factors correlate weakly or not at all. Formally, the response of a respondent to item j is modelled as a weighted sum of the latent factors,

$$x_j = \lambda_{j1}f_1 + \lambda_{j2}f_2 + \cdots + \lambda_{jm}f_m + \varepsilon_j, \quad (2)$$

where $f_1, \dots, f_m$ are the latent factors, the *loadings* λ_{jk} quantify how strongly item j depends on factor k , and ε_j is the part of the item unique to itself (specific variance plus measurement error). The squared loadings summed across factors give the item’s *communality*, the proportion of its variance explained by the common factors.

Exploratory factor analysis (EFA) works backwards from the observed correlation matrix to estimates of the loadings and the number of factors, without assuming in advance which items load on which factor (Costello & Osborne, 2005; Fabrigar et al., 1999). It is “exploratory” in contrast to confirmatory factor analysis, where the loading pattern is specified ahead of time and tested. EFA matters for this thesis in two ways. The *generative* use comes first: the simulation studies of later chapters create artificial questionnaires from a known factor model of the form in Equation 2, so that the ground-truth structure is available for evaluation. The *analytic* use comes second: ReReReRe relies on the fact that attentive respondents reproduce the sample’s factor structure in their own answers, and one variant of the method uses EFA to choose which item pairs to score.

A practical question in any factor analysis is *how many factors to retain*. Retaining too few collapses distinct constructs; too many fits noise. Two classical heuristics are Cattell (1966)’s *scree test*, which looks for an “elbow” in the plot of successive eigenvalues, and the eigenvalue-greater-than-one rule. Both are now regarded as unreliable, and the recommended procedure is *parallel analysis* (Horn, 1965): one generates many random data sets with the same number of variables and respondents but no real factor structure, computes their eigenvalues, and retains only the factors whose eigenvalues in the real data exceed what random data produce by chance. Parallel analysis is, in effect, a permutation-style calibration of the factor count—a recurring logic in this thesis—and is the factor-retention method ReReReRe uses internally when it needs an estimate of dimensionality.

Reliability and Internal Consistency

A measurement is *reliable* to the extent that it is reproducible—that repeated measurement of the same unit yields similar values. For a multi-item scale, the most widely reported reliability index is *coefficient alpha* (Cronbach, 1951), which summarizes the average inter-item correlation, adjusted for the number of items, into a single number between 0 and 1. High alpha indicates that the items “hang together,” covarying as they should if they measure a common construct.

Reliability connects directly to careless responding. Because careless answers are unrelated to the construct, they behave like added noise: they raise the ε_j term in Equation 2, lower communalities, weaken inter-item correlations, and therefore *depress* reliability (Huang et al., 2012). A drop in alpha when a scale is administered to a careless-contaminated sample is one of the symptoms motivating detection in the first place. Conversely, identifying and removing careless respondents typically *raises* the reliability and

sharpens the factor structure of the remaining data—the practical payoff that Chapter 5 evaluates.

Permutation and Randomization Tests

Classical hypothesis tests compare an observed statistic to a theoretical null distribution—the t , F , or χ^2 distribution—derived under assumptions such as normality. When those assumptions are doubtful, a *permutation* (or *randomization*) *test* offers an alternative that builds the null distribution *from the data themselves* (Good, 2005). The logic is simple and assumption-light: if the null hypothesis is true, then the observed arrangement of the data is just one of many equally likely arrangements obtainable by shuffling. One therefore repeatedly shuffles the data in a way consistent with the null, recomputes the statistic on each shuffle, and reads off where the observed value falls in the resulting distribution. A value far in the tail is unlikely under the null and constitutes evidence against it.

This is precisely the inferential principle behind ReReReRe. The method asks, of each respondent, whether their coherence across the item pairs that the sample deems related is higher than would be expected if their answers carried no such structure. The “no structure” null is realized by computing the same coherence over *randomly chosen* item pairs, repeated many times, to form a per-respondent null distribution. The respondent’s observed coherence is then expressed as a *z-score* relative to this distribution,

$$z = \frac{C_{\text{obs}} - \text{mean}(C_{\text{rand}})}{\text{sd}(C_{\text{rand}})}, \quad (3)$$

where C_{obs} is coherence over the genuinely coupled pairs and C_{rand} is the coherence over random pairs. A high z means the respondent is far more coherent than chance—attentive; a low or negative z means they are no more coherent than random pairs would predict—potentially careless. Building the baseline per respondent makes the method *self-calibrating*: it does not assume any particular distribution for the coherence statistic, sidestepping exactly the normality problems that, as Chapter 1 noted, make the chi-square cutoff for Mahalanobis distance unreliable on Likert data.

The Bootstrap

A close cousin of the permutation test is the *bootstrap* (Efron, 1979; Efron & Tibshirani, 1993), a resampling method for quantifying the uncertainty of an estimate. Where a permutation test shuffles to simulate a null hypothesis, the bootstrap *resamples with replacement* to simulate the sampling variability of a statistic. Given a sample of n units, one draws many new samples of size n by sampling the original units with replacement (so some appear multiple times and others not at all), recomputes the statistic of interest on each resample, and uses the spread of these recomputed values as an estimate of the statistic’s sampling distribution.

The bootstrap is valuable precisely when a statistic is too complicated for a closed-form standard error. This thesis uses it for exactly such a case. To quantify how much ReReReRe contributes to the ensemble, Chapter 4 reports the *difference* in Matthews correlation coefficient between the full ensemble and an ablated ensemble without ReReReRe.

No textbook formula gives the standard error of a difference of two MCCs, so a *percentile bootstrap* confidence interval is used: respondents are resampled with replacement several hundred times, the difference is recomputed on each resample, and the 2.5th and 97.5th percentiles of the resulting distribution form a 95% interval. If that interval excludes zero, the contribution is statistically distinguishable from no contribution—the form in which the thesis’s central ablation claim is stated.

Supervised Classification and the Random Forest

The ensemble at the heart of Chapters 4 and 5 is a *supervised classifier*: a model trained on examples whose true labels (careless or attentive) are known, which then predicts labels for new respondents. The specific classifier is the *Random Forest* (Breiman, 2001), built from *decision trees*. Because many psychologists know logistic regression but not tree ensembles, this section develops the Random Forest from the ground up; a fuller treatment appears in standard references (Hastie et al., 2009; James et al., 2013).

Decision trees

A *decision tree* is a sequence of yes/no questions about the predictor variables, arranged as a branching flowchart (Breiman et al., 1984). To classify a respondent, one starts at the top and answers each question—“is the ReReReRe z-score below -0.5 ?”, then “is the longstring above 12?”—following the corresponding branch until reaching a terminal *leaf*, which carries a predicted class (or class probability). The tree is *grown* from training data by repeatedly choosing, at each node, the question that best separates careless from attentive respondents according to a purity criterion such as the Gini index. A single tree has an appealing property—its decisions are transparent and can interact predictors automatically (the second question asked depends on the answer to the first)—but also a serious flaw: it is *high-variance*. Small changes in the training data can change which question is chosen near the top and thereby reshape the entire tree.

From one tree to a forest

The Random Forest tames this instability by averaging *many* de-correlated trees (Breiman, 2001). Two sources of randomness make the trees differ. First, each tree is grown on a *bootstrap resample* of the training respondents (the resampling idea of Section), so each tree sees a slightly different data set—a strategy called *bagging*, for bootstrap aggregating. Second, at each node only a *random subset* of the predictors is considered as candidate questions (typically about $\sqrt{p}$ of the p predictors), which prevents every tree from being dominated by the same one or two strong predictors. The forest’s prediction is the average of the predictions of its trees—in our setting, the proportion of trees that vote “careless,” interpreted as a predicted careless probability. Averaging many noisy but roughly unbiased trees reduces variance without inflating bias, which is why the forest predicts far more stably than any single tree.

Why a forest can beat logistic regression

Logistic regression models the log-odds of the outcome as a *linear, additive* function of the predictors. That is an excellent model when the truth is close to linear and additive,

Table 1

The confusion matrix for careless detection. “Positive” denotes a respondent flagged as careless.

	Truly careless	Truly attentive
Flagged careless	True positive (TP)	False positive (FP)
Flagged attentive	False negative (FN)	True negative (TN)

but careless detection is not such a case. The predictors interact: a respondent with both a very low ReReReRe z-score *and* an unusually low intra-individual response variability is far more suspicious than the additive combination of the two signals would imply, because that conjunction is the signature of straightlining. A Random Forest captures such interactions automatically through the nested structure of its splits, without the analyst having to specify interaction terms in advance. In the simulation studies of later chapters, the forest outperforms logistic regression on every evaluation metric, with the largest margins on the hardest cases (short questionnaires and low careless base rates).

A further benefit is *variable importance*: by measuring how much predictive accuracy degrades when a predictor is randomly permuted, a Random Forest ranks the predictors by their contribution (Strobl et al., 2008). This provides a principled way to ask which detectors carry the signal—an analysis the thesis uses to show that ReReReRe and a small set of auxiliary indices account for most of the ensemble’s power.

Cross-Validation

A classifier evaluated on the same data used to train it will look better than it really is, because it can memorize idiosyncrasies of the training sample that do not generalize—a problem called *optimism* or *overfitting*. *Cross-validation* estimates out-of-sample performance without requiring a separate test set (Kohavi, 1995; Stone, 1974). In *k-fold* cross-validation, the data are partitioned into *k* roughly equal folds (here *k* = 5). The model is trained on *k* − 1 folds and used to predict the held-out fold; this is repeated *k* times so that each fold serves once as the held-out set. Every respondent thus receives a prediction from a model that did *not* see them during training. These *out-of-fold* predictions are concatenated and used for all performance estimates, giving an honest picture of how the classifier would behave on new respondents drawn from the same population. The folds are *stratified*—constructed to preserve the careless base rate within each fold—so that no fold is unrepresentative.

Evaluating a Binary Classifier

Once a classifier assigns each respondent a predicted careless probability, turning those probabilities into a binary decision requires a *threshold*, and judging the decisions requires *metrics*. Both rest on the *confusion matrix* (Table 1), which cross-tabulates the predicted label against the truth into four counts: true positives (TP, careless correctly flagged), true negatives (TN), false positives (FP, attentive respondents wrongly flagged), and false negatives (FN, careless respondents missed).

Table 2*Principal evaluation metrics derived from the confusion matrix.*

Metric	Definition	Range
Sensitivity (recall)	$TP/(TP + FN)$	0 to 1
Specificity	$TN/(TN + FP)$	0 to 1
Precision (PPV)	$TP/(TP + FP)$	0 to 1
F1	harmonic mean of precision and recall	0 to 1
MCC	Equation 4	-1 to +1

From these four counts a family of metrics is derived. *Sensitivity* (recall, true-positive rate) is the proportion of truly careless respondents that are flagged, $TP/(TP + FN)$; *specificity* is the proportion of attentive respondents correctly left unflagged, $TN/(TN + FP)$. *Precision* (positive predictive value) is the proportion of flagged respondents who are genuinely careless, $TP/(TP + FP)$. The *F1 score* is the harmonic mean of precision and recall. These are summarized in Table 2.

The thesis adopts the *Matthews correlation coefficient* (MCC) as its primary metric (Chicco & Jurman, 2020; Matthews, 1975). The MCC is, in effect, the Pearson correlation between the predicted and the true binary labels,

$$MCC = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}, \quad (4)$$

ranging from -1 (perfect disagreement) through 0 (chance) to $+1$ (perfect agreement). Its decisive advantage over accuracy and F1 is that it uses all four cells of the confusion matrix and is therefore robust to *class imbalance*, which careless detection always involves: when only, say, 10% of respondents are careless, a useless classifier that flags no one achieves 90% accuracy, and even F1 can be inflated, whereas the MCC of such a classifier is 0 (Chicco & Jurman, 2020). A high MCC requires the classifier to do well on *both* classes simultaneously.

Two *threshold-free* metrics complete the picture by summarizing performance across all possible thresholds at once. The *area under the receiver operating characteristic curve* (AUC) is the probability that a randomly chosen careless respondent receives a higher predicted probability than a randomly chosen attentive one (Hanley & McNeil, 1982); 0.5 is chance and 1 is perfect. The *area under the precision-recall curve* (AUPRC) is a companion measure that, by ignoring the typically very large true-negative cell, is more informative than AUC under heavy class imbalance (Saito & Rehmsmeier, 2015). Because AUC and AUPRC do not depend on a chosen cutoff, they isolate the quality of the underlying ranking from the separate question of where to set the threshold.

That separate question—*which* threshold to use—is itself consequential, as Chapter 1 stressed. A threshold defines an *operating point*, a specific trade-off between sensitivity and specificity; moving it trades missed careless respondents against falsely flagged attentive ones. Chapter 5 returns to the problem of choosing an operating point without access to ground-truth labels, which is the form the threshold problem takes in real deployment.

Summary

The pieces assembled here recur throughout the thesis as a connected toolkit rather than a list. Correlation, computed at the level of the individual rather than the sample, is what ReReReRe measures; the common-factor model explains why that correlation exists for attentive respondents and parallel analysis estimates its dimensionality; the permutation test turns the measurement into a self-calibrating per-respondent z-score; the bootstrap puts a confidence interval on the detector's contribution; the Random Forest combines that detector with auxiliary indices, cross-validation estimates the combination's performance without optimism, and the Matthews correlation coefficient—together with AUC and AUPRC—reports how good that performance is under the class imbalance intrinsic to careless detection. With these concepts in place, Chapter 3 can introduce the ReReReRe detector itself.

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